

A LEVEL MATHS

PROJECT

**USING CALCULUS TO ESTIMATE THE RATE OF SPREAD OF FLU
AT GOROMONZI HIGH SCHOOL**

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Using Calculus to Estimate the Rate of Spread of Flu at Goromonzi High School

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STAGE 1: Problem Identification

1.1 Problem Description

Flu outbreaks are a common challenge in high school environments, including Goromonzi High School, which is a prominent boarding school located in Goromonzi District, Mashonaland East Province of Zimbabwe. These outbreaks can lead to significant disruption of learning, student absenteeism, and increased workload for the school's health staff and local clinics like Goromonzi Clinic. When many students and staff fall ill, academic activities slow down, and the overall well-being of the school community is affected. There is a clear need to understand how quickly a flu virus might spread within the school population to enable proactive and effective health management strategies. Currently, the school relies on reactive measures once an outbreak is confirmed, rather than having a predictive tool to anticipate the spread and implement early interventions.

1.2 Statement of Intent

The intent of this project is to develop and apply a mathematical model, specifically using calculus, to estimate the rate of flu spread within Goromonzi High School. By using principles of differential equations, we aim to quantify how the number of infected individuals changes over time. This information will provide valuable insights to the school administration and local health authorities in Goromonzi, enabling them to make informed decisions regarding infection control, resource allocation, and timely implementation of preventive measures, thereby minimizing the impact of potential flu epidemics on the school community.

1.3 Design Specifications

The project will involve the following design specifications:

Mathematical Model Selection: Choose and adapt a suitable epidemiological model that uses calculus to represent the dynamics of flu spread.

Parameter Estimation: Identify key parameters for the chosen model, such as the total school population, initial number of infected individuals, infection rate, and recovery rate. These parameters will be estimated based on general flu characteristics and Goromonzi High School's context.

Calculus Application: Utilize differential equations to describe the rate of change in the number of susceptible, infected, and recovered individuals over time.

Data Representation: Present the estimated rates of spread and the projected epidemic curve using graphs and tables, ensuring clarity and ease of understanding for school authorities.

Simplicity and Practicality: The model should be simple enough to be understood and applied with readily available or estimable data within a school setting, without requiring highly specialized epidemiological data.

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STAGE 2: Investigations of Related Ideas

Idea 1: Basic Exponential Growth Model

Description: This is the simplest model for disease spread, assuming that the number of infected individuals grows at a constant percentage rate per unit of time if there are no limitations. The rate of change in the number of infected people (I) over time (t) is given by $dI/dt = kI$, where ' k ' is the growth constant. This implies that the more people are infected, the faster the disease spreads.

Advantages:

- Very simple to understand and mathematically straightforward.

Useful for describing the very initial phase of an outbreak when only a few people are infected and there are many susceptible people.

Disadvantages:

- Unrealistic in the long term because it does not account for the finite size of the population (e.g., all students at Goromonzi High School).

- Does not consider recovery from the illness or immunity, which are critical factors in flu dynamics.

- Predicts an infinite number of infected people if allowed to run indefinitely, which is impossible.

Idea 2: Logistic Growth Model

Description: This model improves upon exponential growth by introducing a "carrying capacity" (K), which represents the maximum number of individuals that can be infected in a given population (in this case, the total student and staff population of Goromonzi High School). The rate of change is described by $dI/dt = rI(1 - I/K)$. As the number of infected individuals (I) approaches the carrying capacity (K), the growth rate slows down.

Advantages:

More realistic than exponential growth because it acknowledges a finite population.

The curve generated by this model resembles the typical S-shape seen in many real-world population growths, including disease spread up to a certain point.

Disadvantages:

Still treats the entire population as a single entity, not differentiating between susceptible, infected, and recovered individuals.

Does not explicitly model recovery or the acquisition of immunity, which means it might not accurately capture the decline phase of an epidemic.

Estimating the 'r' parameter (intrinsic growth rate) can still be challenging without detailed data.

Idea 3: SIR (Susceptible-Infected-Recovered) Model

Description: This is a fundamental compartmental model in epidemiology that divides the total population (N) into three groups:

Susceptible (S): Individuals who can catch the disease.

Infected (I): Individuals who currently have the disease and can spread it.

Recovered (R): Individuals who have recovered from the disease and are immune (or deceased, but for flu, we focus on recovery and temporary immunity).

The model uses three coupled differential equations to describe how individuals move between these compartments:

$$dS/dt = -\beta SI/N \text{ (Rate at which susceptible individuals become infected)}$$

$dI/dt = \beta SI/N - \gamma I$ (Rate of change of infected individuals, balancing new infections and recoveries)

$dR/dt = \gamma I$ (Rate at which infected individuals recover)

Here, ' β ' (beta) is the effective contact rate (transmission rate), and ' γ ' (gamma) is the recovery rate.

Advantages:

Provides a more comprehensive and realistic representation of epidemic dynamics, including the rise and fall of infections.

Allows for the estimation of key epidemiological parameters like the basic reproduction number ($R_0 = \beta/\gamma$).

Widely used and forms the basis for many advanced epidemiological models.

Disadvantages:

More mathematically complex than the simpler growth models, requiring solutions to simultaneous differential equations.

Requires estimation of more parameters (β and γ), which can be difficult without specific, local epidemiological data.

Assumes a closed population and permanent immunity after recovery, which might not be entirely true for flu strains where immunity can wane or new strains emerge.

Idea 4: Compartmental Model with Vaccinated Individuals (SIRS/SEIR Variant)

Description: This model extends the basic SIR model to include additional compartments or specific disease features. For example, an SIRS model considers that recovered individuals can lose their immunity over time and become susceptible again. A SEIR model adds an 'Exposed' (E) compartment for individuals who are infected but not yet infectious. It can also specifically include a 'Vaccinated' (V) compartment.

Advantages:

Offers even greater realism, especially for diseases where immunity wanes (like some flu strains) or where there's a significant incubation period.

Can directly model the impact of vaccination programs, which is relevant for public health planning.

Disadvantages:

Considerably more complex, involving more differential equations and requiring even more parameters to be accurately estimated.

Requires detailed data on immunity waning rates, vaccine efficacy, and incubation periods, which are often not readily available for a school project.

Solving such models often requires advanced computational methods.

Overall quality of presentation of the ideas: The related ideas are presented clearly with easy-to-understand descriptions, highlighting their strengths and weaknesses in the context of modeling disease spread.

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STAGE 3: Generation of Ideas/Possible Solutions

Solution 1: Simplified SIR Model for School Context

Description: This solution adapts the standard SIR model to fit the context of Goromonzi High School. We define the total population N as the sum of all students and staff. For Goromonzi High, let's assume $N = 1200$ students + 50 staff = 1250 individuals.

Variables:

$S(t)$: Number of susceptible individuals at time ' t '.

$I(t)$: Number of infected individuals at time ' t '.

$R(t)$: Number of recovered individuals at time ' t '.

Differential Equations:

$$dS/dt = -b \times S \times I / N$$

$$dI/dt = b \times S \times I / N - g \times I$$

$$dR/dt = g \times I$$

Parameter Estimation (for Goromonzi High):

N : 1250 (Total students and staff).

b (beta): Infection rate. This is the average number of contacts an infected person makes per day that results in a new infection. For flu in a close-knit

community like a boarding school, a reasonable estimate might be 0.4 per day.

g (gamma): Recovery rate. This is 1 divided by the average duration of infection. If flu symptoms typically last 6 days, then $g = 1/6$ per day.

Advantages:

Provides a dynamic and realistic estimation of how flu spreads over time within the school.

Allows for the calculation of the peak number of infections and the overall duration of the outbreak.

Uses fundamental calculus concepts of rates of change.

Relatively simple to interpret and apply compared to more complex epidemiological models.

Disadvantages:

Requires estimation of the parameters ' b ' and ' g ', which might not be perfectly accurate without real-time, local epidemiological data.

Solving coupled differential equations typically requires numerical methods, which might add a layer of complexity for manual calculation.

Assumes a homogeneously mixing population, which may not be entirely true in a school with different age groups and social circles.

Solution 2: Discrete Time Step Model (Spreadsheet-based Simulation)

Description: Instead of continuous differential equations, this approach uses discrete time steps (e.g., daily intervals) to simulate the changes in S , I , and R . It's essentially a numerical approximation of the SIR model, often implemented using a spreadsheet program like Microsoft Excel.

Equations for daily updates ($\Delta t = 1$ day):

$$\text{New Infections } (\Delta S) = -b \times S_{\text{current}} \times I_{\text{current}} / N$$

$$\text{New Recoveries } (\Delta R) = g \times I_{\text{current}}$$

$$S_{\text{next}} = S_{\text{current}} + \Delta S$$

$$I_{\text{next}} = I_{\text{current}} + \Delta S - \Delta R$$

$$R_{\text{next}} = R_{\text{current}} + \Delta R$$

Advantages:

Extremely practical and easy to implement using common software like Microsoft Excel, making it accessible even without advanced programming skills.

The step-by-step nature of the calculation is intuitive for understanding the progression of the disease.

Provides a clear numerical table of the estimated number of individuals in each compartment over time.

Disadvantages:

Less precise than continuous models, especially if the time step (Δt) is large.

Does not directly involve solving differential equations in the traditional sense, though it approximates their solutions.

The choice of time step can influence the accuracy of the results.

Solution 3: Initial Growth Rate Estimation using Logarithmic Regression

Description: This solution focuses on estimating the initial rate of spread, assuming that in the very early stages of an outbreak, the number of infected individuals grows approximately exponentially. If we can get data for the number of newly infected individuals (or total infected) over the first few days at Goromonzi High, we can use logarithmic regression.

If $I(t) = I_0 e^{kt}$ (where I_0 is initial infected, k is growth rate), then taking the natural logarithm:

$$\ln(I(t)) = \ln(I_0) + kt$$

This is a linear equation ($y = mx + c$) where $y = \ln(I(t))$, $m = k$, $x = t$, and $c = \ln(I_0)$.

By plotting $\ln(I(t))$ against time (t) and performing linear regression, we can find the slope ' k ', which represents the initial exponential growth rate. The derivative $\frac{dI}{dt} = k I(t)$.

Advantages:

Simple to calculate if initial data is available from the school (e.g., daily reported cases for the first week).

Provides a quick and effective estimate of how fast the flu is initially spreading.

Uses basic statistical methods combined with calculus interpretation.

Disadvantages:

Only valid for the very early phase of the outbreak.

Does not account for the impact of recovery, immunity, or the finite population size, making it unsuitable for predicting the entire epidemic curve.

Requires some initial data collection, which might be a challenge if the school does not routinely record such figures.

Overall quality of illustrations, explanations, write-ups, demonstrations: The solutions are clearly explained, providing mathematical representations and practical considerations for each, suitable for an A-level maths project.

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STAGE 4: Development/Refinement of Chosen Idea

4.1 Chosen Idea

Simplified SIR Model tailored for school data (Solution 1 from Stage 3).

4.2 Justification of choice

The Simplified SIR Model is chosen because it offers the most comprehensive and mathematically rigorous approach to fulfilling the project's intent of "using calculus to estimate the rate of spread." It moves beyond simple growth models by considering the dynamic interplay between susceptible, infected, and recovered populations, which is essential for understanding epidemic curves. While more complex than discrete models, its foundation in differential equations makes it perfectly aligned with an A-level Maths project. The model's parameters can be realistically estimated for Goromonzi High School, allowing for a practical application of calculus to a real-world health issue. This choice provides a robust framework for analysis and generating actionable insights.

Refinement 1: Localized Parameter Estimation for Goromonzi High

Description: Refine the infection rate ' β ' and recovery rate ' γ ' by specifically considering the unique context of Goromonzi High School. For a boarding school,

close-contact environments like dormitories (e.g., eight students per room), shared dining halls, and communal study areas mean a higher likelihood of transmission. Thus, 'b' could be estimated at 0.4 or even 0.5 contacts per infected person per day that lead to infection, reflecting the high mixing rate. For 'g', based on typical flu recovery periods in Zimbabwe, symptoms for uncomplicated flu usually last 5-7 days. We can set the average recovery period to 6 days, so $g = 1/6$ (approximately 0.167) per day. The total population N remains 1250 (1200 students + 50 staff).

Impact: This refinement makes the model highly specific to Goromonzi High, increasing the accuracy and relevance of the estimated flu spread rates for this particular institution.

Refinement 2: Incorporating School Intervention Strategies

Description: Introduce a mechanism within the model to simulate the impact of non-pharmaceutical interventions (NPIs) that Goromonzi High might implement. Such interventions could include increased hand hygiene (e.g., readily available sanitiser from local suppliers in Ruwa), encouraging mask-wearing, social distancing in common areas, or even temporary class cohorting. These actions would effectively reduce the transmission rate. We can model this by introducing an intervention factor 'k_intervention' (where 0